

Continuous functions – 6

Functional equations under a regularity assumption

Homework

1 сентября 2026 года

Theoretical minimum. The Cauchy equation and the regularity conditions that force linearity; Hamel bases and pathological solutions; the exponential, logarithmic, power and Jensen equations; d'Alembert's equation $f(x+y) + f(x-y) = 2f(x)f(y)$; iterative equations $f(f(x)) = g(x)$; the trick of proving an identity first on $\mathbb{Z}$, then on $\mathbb{Q}$, then by continuity on $\mathbb{R}$.

1. Prove that an additive function bounded above on a set of positive measure is linear, and deduce the measurable version of Cauchy's theorem.
2. Find all continuous $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(x+y) + f(x-y) = 2f(x)f(y)$ (d'Alembert), and describe the solutions without the continuity assumption.
3. Find all continuous $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(x+y) = f(x) + f(y) + f(x)f(y)$.
4. Prove that a continuous f with $f(f(x)) = x$ for all x is either the identity or a decreasing involution, and describe all such involutions.
5. Find all continuous $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying $f(2x) = 2f(x)$ and $f(x) \leq x$ for all x , and show that dropping continuity produces other solutions.
6. Prove that the only continuous solutions of $f(x)f(y) = f(xy)$ on $(0, \infty)$ which are bounded on $[1, 2]$ are the powers x^c , and give a discontinuous solution built from a Hamel basis.
7. (Schröder's equation) Let g be continuous with $g(0) = 0$ and $0 < g'(0) < 1$. Prove that the functional equation $\varphi(g(x)) = g'(0)\varphi(x)$ has a continuous solution near 0 with $\varphi'(0) = 1$, and explain why it linearises the iterations of g .

8. Find all continuous $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) + f\left(\frac{1}{1-x}\right) = x$ for all $x \neq 0, 1$, using the fact that $x \mapsto \frac{1}{1-x}$ has order three.
9. Prove that a continuous $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(x+1) = f(x)$ and $f(2x) = f(x)$ is constant.
10. (RMO) Find all continuous $f: [0, 1] \rightarrow \mathbb{R}$ such that $\int_0^1 f\left(\sqrt[n]{x}\right)^{n-k} dx = \frac{k}{n}$ for every $k \in \{1, \dots, n-1\}$, where $n \geq 3$ is a fixed odd integer.